Amit Manicka

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PhD student in Mechanical Engineering at the University of Minnesota researching medical robotics and devices, with an MS in Mechanical Engineering and a second MS in Electrical Engineering in progress (analog circuit design, microelectronics, mechatronics). Five years at Calyan Technologies spanning mechanical, electrical, firmware, and software engineering for implantable pacemakers.

Education

University of Minnesota – Twin Cities Minneapolis, MN

Doctor of Philosophy (PhD), Mechanical Engineering *Fall 2025 – Present (expected 2029)*

- Focus: medical robotics and devices, computer vision, simulation, and ML for biological decision-making. Advisor: Prof. Tim Kowalewski, Medical Robotics and Devices (MRD) Laboratory.

Master of Science (MS), Electrical Engineering *Fall 2025 – Present (expected Dec 2027)*

- Specialization: analog circuit design, microelectronics, and mechatronics.

Master of Science (MS), Mechanical Engineering *Fall 2024 – Spring 2025*

- GPA 3.869/4.00. Integrated MSME in robotics, controls, and automation; research on impedance control for insertion.

BS Computer Science & Bachelor of Mechanical Engineering (dual degrees) *Fall 2021 – Summer 2024*

- GPA 3.954/4.00 (graduate coursework 4.00/4.00). Completed both degrees in three years. University Honors Program, Dean's List every semester, Iron Range Scholarship, U of MN Presidential Scholarship.

Selected graduate coursework: Medical Robotics · Advanced Controls Design · Feedback Controls · Linear Systems & Controls · Optimal Estimation · Intro. to RF & Microwave Engineering · Finite Element Analysis · Applied Stress Analysis · Computer Vision · Applied Machine Learning · Machine Learning Fundamentals · Robotics (CS) · Computational Matrix Theory · New Product Design & Business Development I/II (Medtronic team)

Research Experience

Medical Robotics and Devices (MRD) Laboratory, University of Minnesota Minneapolis, MN
Graduate Researcher (PhD), Dept. of Mechanical Engineering — PI: Prof. Tim Kowalewski *Summer 2026 – Present*

- Doctoral research on medical robots and devices that use advanced decision-making to identify and interact with complex human anatomy, combining computer vision, physics simulation (NVIDIA Isaac for Healthcare), and ML.

Donath Robotics Laboratory, University of Minnesota Minneapolis, MN
Graduate Researcher, Dept. of Mechanical Engineering — PI: Prof. Max Donath *Summer 2024 – Spring 2025*

- Automated latching/unlatching of electrical and pneumatic brake lines on an 18-wheel truck with a single UR5 arm and two-finger gripper, a task that normally requires a human operator's two hands and ten fingers (Python/C++).
- Implemented impedance control to modulate contact forces, protecting the latching mechanism and eliminating arm jerk; used computer vision so the system retrofits any truck instead of redesigning the truck around the robot.
- Designed and oversaw fabrication of all development and test fixtures; wrote the grant proposals that funded the work.
- Protocol presented to Amazon for sponsored implementation in its trucking and shipping operations.

Tithof Laboratory, University of Minnesota Minneapolis, MN
Undergraduate Researcher, Dept. of Mechanical Engineering — PI: Prof. Jeff Tithof *Fall 2023 – Spring 2024*

- Awarded a UROP grant to build capillary-scale models of the mouse glymphatic system; augmented cerebrospinal-fluid (CSF) flow models with perivascular capillary density to study solute transport and waste clearance (MATLAB/Python).
- Developed 3D simulations of CSF and interstitial-fluid exit pathways at varying brain depths, with applications to Alzheimer's, Parkinson's, and sleep research; presented at the 2024 Spring Undergraduate Research Symposium.

Robotics, Perception and Manipulation (RPM) Laboratory, Minnesota Robotics Institute Minneapolis, MN
Undergraduate Researcher, Dept. of Computer Science — PI: Prof. Karthik Desingh *Spring 2023 – Summer 2023*

- Developed Python navigation algorithms for Boston Dynamics Spot to traverse tight spaces, detect and relocate obstacles in its path using computer vision, and record and revisit prior locations via graph-based path planning.

Complex Fluids and Multiphase Flows Laboratory, University of Minnesota Minneapolis, MN
Undergraduate Researcher, Dept. of Mechanical Engineering — PI: Prof. Cari Dutcher *Fall 2022 – Spring 2023*

- Ran traveling-surface-acoustic-wave sorting experiments and analyzed particle displacement vs. size and voltage toward a type 1 diabetes cell therapy; wrote Python tools characterizing the acoustic radiation force factor. Co-author on the resulting preprint (arXiv:2509.09692).

Professional Experience

Calyan Technologies Oakdale, MN
Mechanical, Electrical & Software Engineer (2025 – Present); Engineering Intern (2021 – 2025) *Summer 2021 – Present*

Electrical Engineering

- Designed opcode decoders and worked with I^2C and UART communication for implantable pacemaker electronics.
- Developed autonomous hardware-in-the-loop (HIL) live-testing platforms for continuous validation of device firmware and pacemaker/digital-twin communication.
- Devised and built the electrical schematic and fixtures for fatigue testing of experimental pacemaker antennas; built baseline circuit boards for electrical-system validation; extensive hand soldering and prototype board assembly.

Mechanical Engineering

- Mechanical design verification of a VVIR pacemaker and its implant instrumentation; optimized the mechanical attachment clip in SOLIDWORKS and performed stress and fatigue FEA on pacemaker components in ANSYS.
- Designed and executed fatigue test protocols for the flex arm and Bluetooth antenna (pending FDA review), plus mechanical failure testing of pacemaker circuitry and battery-efficiency analysis.

Software & Firmware Engineering

- Wrote Bluetooth firmware linking the pacemaker to an Android tablet programmer; tested EGM and BLE links end-to-end.
- Developed a patient mobile application for pacemaker interfacing and care-team communication, and a physician-facing monitoring application.
- Built internet gateways and a server for a central patient-data collation site enabling algorithmic patient management and AI applications; performed anomaly and vulnerability detection on device software and remote monitoring.

Medtronic — Student Consultant, Cardiac Rhythm Management Minneapolis, MN
U of MN New Product Design & Business Development Program (ME 8221/8222) *Fall 2025 – Spring 2026*

- Nine-month consulting engagement under Drs. Alexander Mattson, Zhongping Yang, and Lillian Kornet and Victoria Schrimpf: designed and marketed novel delivery systems for cardiac rhythm management devices.
- Owned the front-end process end to end: voice-of-customer interviews, patent and competitive analysis, paper and physical prototyping, CAD modeling, and promotional video for stakeholder review.

Teaching Experience

ME 5286 – Robotics, University of Minnesota Minneapolis, MN
Graduate Teaching Assistant (two appointments) *Fall 2024 – Summer 2025 & Spring 2026*

- TA for the graduate hands-on robotics course (Prof. Max Donath, 2024–25; Prof. Rachel Gelhar Humann, 2026) covering forward/inverse kinematics, joint configurations, robot programming, and computer vision; managed the other TAs.
- Prepared and ran UR5 laboratories: range of motion, precision and repeatability measurement, rapid flashlight assembly with a Robotiq 2-finger gripper, and vision/ML-based sorting of hardware tools.
- Built Arduino-based force/torque DAQ hardware to measure UR5 accuracy and repeatability; redesigned the gripper force sensor for impedance control; integrated new force-torque sensors and repaired robot joints.

Publications & Presentations

- N. Sethia, J. S. Rao, **A. Manicka**, M. L. Etheridge, E. B. Finger, J. Bischof, and C. S. Dutcher (2025). "Image detection–based high-throughput sorting of particles using traveling surface acoustic waves in microscale flows." *arXiv* preprint, arXiv:2509.09692.
- **A. Manicka** (2024). "Using High-Resolution Mouse Vasculature Data to Develop Realistic Simulations of Brain Waste Clearance." Spring Undergraduate Research Symposium, University of Minnesota. University Digital Conservancy, hdl.handle.net/11299/265719.

Selected Projects

- **7-DoF Delta 3D Printer** (Bakken Medical Devices Center): firmware and UI for printing stents and prototype implants; designed a removable print bed, repaired the extruder, and mentored an undergraduate through the mechanical work.
- **Bariatric Standing Aide:** CAD design, FEA, and prototype of a mobility-rehabilitation device for patients up to 700 lb.

Technical Skills

- **Programming:** Python, C/C++, Java, MATLAB, JavaScript, OCaml, x86 Assembly, NumPy, Android development
- **Robotics & Controls:** UR5, Boston Dynamics Spot, Robotiq grippers, RoboDK, impedance/feedback/real-time control, kinematics, computer vision, machine learning, NVIDIA Isaac for Healthcare
- **Electrical & Embedded:** analog circuit design, opcode decoders, Arduino, firmware, hardware-in-the-loop test platforms, KiCad, QUCS, soldering and board assembly; working familiarity with I^2C, UART, and BLE
- **Mechanical & Fabrication:** SOLIDWORKS, PTC Creo, ANSYS FEA, stress/fatigue testing, failure analysis, design verification, rapid prototyping, grant writing